

Vraj Shah

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Education

The Ohio State University

Bachelor of Science, Computer Science & Engineering, Columbus, OH

Graduation: May 2025

Work Experience

Research Assistant

August 2024 – Present

Machine Analysis of Political Speech Lab, Columbus, OH

- Developed and optimized a scalable data pipeline leveraging Computer Vision and NLP to analyze political speech, with an SQL database cataloging politician profiles, topics, and video segments.
- Automated data extraction using Python, WhisperX, and Meta's Llama-3, reducing debate processing time to 13 minutes, with data visualizations created via matplotlib.

Backend Engineer

June 2024 – Present

dustin, Columbus, OH

- Developed "dustin," an AI-driven platform for e-commerce sellers, integrating expense tracking and demand forecasting, securing our first paying customer.
- Implemented key technologies including Plaid API, Next.js, LangChain, and Supabase for seamless financial data management and application scalability.

IT Support Technician

May 2024 – August 2024

Digital Union Classroom Services, Columbus, OH

- Provided comprehensive customer service and technical support in classrooms and Digital Union Studios, assisting with instructional technology, software, and AV systems, while performing daily maintenance and troubleshooting.

Full-Stack Developer

December 2022 – May 2024

Mindful Diagnosis, Columbus, OH

- Led a non-profit startup & developed an implicit bias assessment platform for physicians using NextJS, SQL, Mux, Tailwind, and Prisma, securing \$1500 in startup funding from Ohio State University.

Research Assistant

June 2022 – August 2022

Network-Based Computing Lab, Columbus, OH

- Optimized the NOWLAB website with HTML/CSS and GitHub version control.
- Configured network protocols & created infrastructure for lab hardware.

Projects and Awards

National Cyber League

October 2018 – Present

- Cybersecurity Capture the Flag (CTF) competition, honing skills in log analysis, binary exploitation, open-source intelligence, cryptography, etc.
- Received Platinum award for being placed 100th out of 5380 individuals from colleges around America.

Gym Data Tracking App Project

May 2024 - August 2024

- Developed a real-time gym tracking application for Ohio State University gyms using React Native and Selenium to scrape and display gym and station usage data, and successfully submitted the app to the Apple App Store for accessibility by OSU students and staff.

HackOH/IO Makeathon

March 2024

- Developed an advanced security suite, using machine learning for facial recognition to unlock doors, integrated with Twilio API for alerts on unknown entries, utilized weather API for real-time updates, and implemented an audio level detector for gunshot detection alerts via SMS.
- Placed 2nd out of 100 competing teams.

HackOH/IO Hackathon

October 2022

- Developed a safety application within a 24-hour timeframe that utilizes Google's API for location tracking to identify and display the nearest emergency blue light from the user's current location.
- Placed 2nd out of 200 competing teams.

Skills

Languages: English (Native), Gujarati (Native), Hindi (Second)

Programming Languages: Java, C++, C, x86-64 Assembly, Python, TypeScript, React Native, JavaScript (ThreeJS, VueJS, NextJS).

Technical Skills: Linux, HTML5, CSS, Git, Microsoft Suite, SQL, Bash, SolidWorks, Visual Studio, Figma, Twilio, WebGL, OpenCV, Tailwind CSS, Mux, SDLS, Rest APIs, WhisperX, NLP